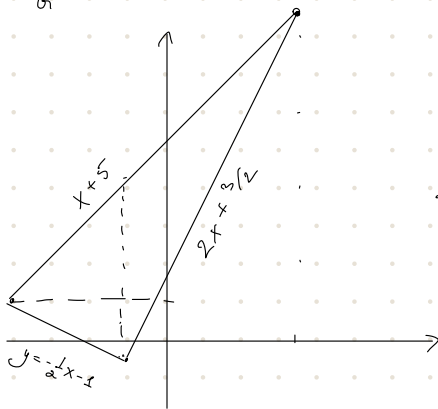


b) $\iint_G \sin \pi(x-y) dx dy$, $G = \Delta: \begin{pmatrix} -4, 1 \\ -1; -\frac{1}{2} \\ 7/2; 17/2 \end{pmatrix}$



$$y = kx + b$$

$$\begin{cases} 1 = -4k + b \\ -\frac{1}{2} = -k + b \end{cases}$$

$$\begin{cases} -\frac{1}{2} = -k + b \\ \frac{17}{2} = \frac{7}{2}k + b \end{cases}$$

$$-\frac{3}{2} = 3k$$

$$\frac{18}{2} = \frac{9}{2}k$$

$$-\frac{1}{2} = k; b = -1$$

$$y = -\frac{1}{2}x - 1$$

$$\begin{cases} k = 2 \\ b = -\frac{1}{2} + 2 = \frac{3}{2} \end{cases}$$

$$\begin{cases} 1 = -4k + b \\ \frac{17}{2} = \frac{7}{2}k + b \end{cases}$$

$$\frac{15}{2} = \frac{15}{2}k \quad k = 1;$$

$$b = 1 + 4 = 5$$

$$y = x + 5$$

$$\iint_G \sin \pi(x-y) dx dy = \int_{-4}^{-1} dx \int_{-\frac{1}{2}x-1}^{x+5} \sin \pi(x-y) dy + \int_{-1}^{7/2} dx \int_{2x+\frac{3}{2}}^{x+5} \sin \pi(x-y) dy$$

$$= \left\{ \begin{array}{l} \sin \pi(x-y) dy \\ d(\pi(x-y)) = -\pi dy \end{array} \right. = +\frac{1}{\pi} \cos(\pi(x-y)) \left\{ \right.$$

$$= \int_{-4}^{-1} \left[\frac{1}{\pi} \cos(\pi(x-y)) \right]_{-\frac{1}{2}x-1}^{x+5} dx + \int_{-1}^{7/2} \left[\frac{1}{\pi} \cos(\pi(x-y)) \right]_{2x+\frac{3}{2}}^{x+5} dx =$$

$$= \int_{-4}^{-1} \left(\frac{1}{\pi} (\cos \pi(x-x-5)) - \cos(\pi(x+\frac{1}{2}x+1)) \right) dx +$$

$$+ \int_{-1}^{7/2} \left(\frac{1}{\pi} (\cos(\pi(x-x-5)) - \cos(\pi(x-2x-\frac{3}{2}))) \right) dx =$$

$$= \int_{-4}^{-1} \frac{1}{\pi} (-1 - \cos(\pi \cdot \frac{3}{2}x + \pi)) dx - \int_{-1}^{7/2} \frac{1}{\pi} (-1 - \cos(-\pi x - \frac{3}{2}\pi)) dx =$$

$$= \int_{-4}^{-1}$$

② Выведем.

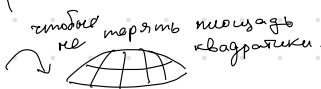
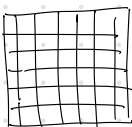
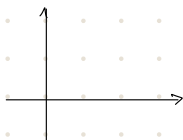
$$\iint_D \sqrt{4-x^2-y^2} \, dx \, dy, \quad X: x^2+y^2 \leq 2x$$

$$\iint_D f(x(u,v), y(u,v)) \cdot |J(x)| \, dx \, dy$$

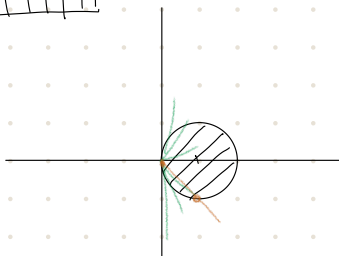
$$\begin{cases} x = x(u,v) \\ y = y(u,v) \end{cases}$$

$$\int f(\varphi(t)) \varphi'(t) \, dt$$

$$J(x) = \begin{vmatrix} x'_u & x'_v \\ y'_u & y'_v \end{vmatrix}$$



$$\begin{cases} x = r \cdot \cos \varphi \\ y = r \sin \varphi \end{cases}$$



$$\begin{aligned} x^2 + y^2 - 2x &\leq 0 \\ (x^2 - 2x + 1) - 1 + y^2 &\leq 0 \\ (x-1)^2 + y^2 &\leq 1 \end{aligned}$$

$$\sqrt{4-x^2-y^2} = \sqrt{4-r^2\cos^2\varphi - r^2\sin^2\varphi} = \sqrt{4-r^2(\cos^2\varphi + \sin^2\varphi)} = \sqrt{4-r^2}$$

$$J = \begin{vmatrix} \cos \varphi & -r \sin \varphi \\ \sin \varphi & r \cos \varphi \end{vmatrix} = r \cos^2 \varphi + r \sin^2 \varphi = r(\cos^2 \varphi + \sin^2 \varphi) = r$$

↑
мы и r

$$D: \quad 0 \leq r \leq 2 \quad -\frac{\pi}{2} \leq \varphi < \frac{\pi}{2}$$

$$\begin{aligned} \int_{-\pi/2}^{\pi/2} d\varphi \int_0^{2\cos\varphi} \sqrt{4-r^2} \, r \, dr &= \left\{ d(4-r^2) = -2r \, dr \right\} = \int_{-\pi/2}^{\pi/2} d\varphi \left(\frac{1}{2} \int_0^{2\cos\varphi} \sqrt{4-r^2} \, d(4-r^2) \right) = \\ &= \int_{-\pi/2}^{\pi/2} -\frac{1}{2} \frac{(4-r^2)^{3/2}}{3} d\varphi \Big|_0^{2\cos\varphi} = -\frac{1}{3} \int_{-\pi/2}^{\pi/2} (4-r^2)^{3/2} \Big|_0^{2\cos\varphi} d\varphi = -\frac{1}{3} \int_{-\pi/2}^{\pi/2} (4-4\cos^2\varphi)^{3/2} - 4^{3/2} d\varphi = \end{aligned}$$

$$r^2 \cos^2 \varphi + r^2 \sin^2 \varphi \leq r \cdot 2 \cos \varphi \quad ; \quad r \leq 2 \cos \varphi$$

$$= -\frac{1}{3} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (4 - 4 \cos^2 \varphi)^{3/2} d\varphi + \frac{1}{3} \cdot 4^{3/2} \varphi \Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}} = -\frac{1}{3} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 4^{3/2} (1 - \cos^2 \varphi)^{3/2} d\varphi + \frac{8}{3} \pi =$$

$$= -\frac{8}{3} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^3 \varphi d\varphi + \frac{8}{3} \pi = \left[\dots \right] = \dots$$